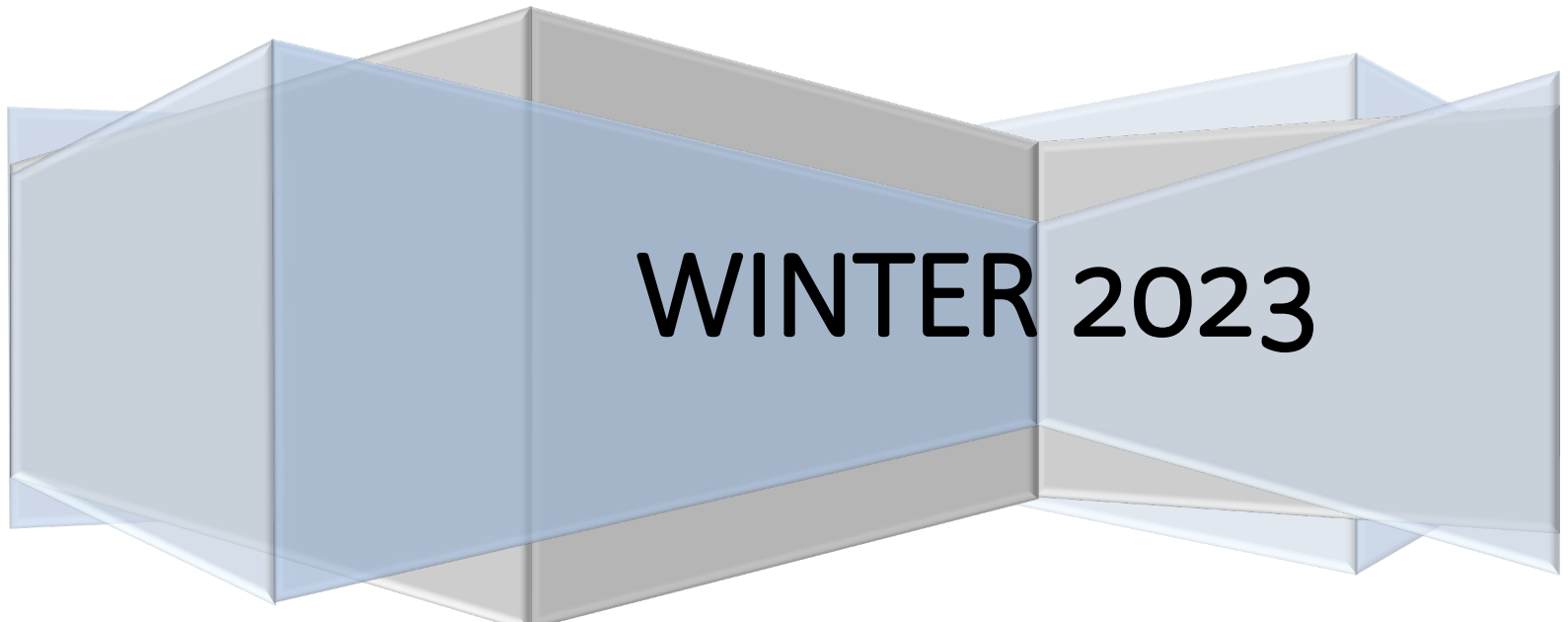


UNIVERSITY OF WOLLONGONG IN DUBAI

FACULTY OF ENGINEERING AND INFORMATION SCIENCES
UG DEGREE PLANNER
HANDOUT VERSION



WINTER 2023

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BACHELOR OF COMPUTER SCIENCE

Testamur Title of Degree:	Bachelor of Computer Science
Abbreviation:	BCompSc
Majors	Cyber Security Game and Mobile Development Big Data
UOWD Course Code:	3165
Total Credit Points:	144 (plus additional GEC requirement subjects)
Duration:	3 years (8 sessions) full time
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

Computer science is the study of algorithmic processes that describe and transform information: theory, analysis, programming and design. The Bachelor of Computer Science focuses on the development of high-level programming skills that can be applied across a wide range of applications, including analysis of stock market trends, games design, neural network design, automatic teller machines and patient monitoring in hospitals. The core subjects help students understand the structure of data and the role it plays in delivering solutions to complex problems. The degree includes core of programming subjects as well as electives in languages, artificial intelligence, computer security, computer graphics, operating systems, real-time software and software engineering. This program provides students with a strong core of computing and allows them to pursue their interest by choosing among many specialised electives or to major in areas such as Cyber Security, Game and Mobile Development, and Big Data.

Degree Requirements

To qualify for award of the degree of Bachelor of Computer Science,

- An aggregate of at least 144cp which includes 90cp from the program core, a 12cp final-year project and 42cp of General Electives as specified in the table below.
- Students need to choose 42cp of electives (7 subjects) if they are not opting for any major of study.
- Students must complete 18cp (3 subjects) 300 level CSCI subjects and not more than 12cp (2 subjects) at 100 level.
- The completion of General Education Components as listed in the table.
- an overall weighted average mark (WAM) of 50%.

Learning Outcomes

Upon successful completion of the Bachelor of Computer Science program, students will be able to:

1. Apply theoretical basis of computer science to solve a range of practical problems
2. Design and develop state of the art software solutions for a variety of applications.
3. Design, develop, and employ effective and efficient approaches and algorithms in solving practical problems.
4. Manage all aspects of the project including analyses of requirements, design of solutions, selection of appropriate tools for implementation, or integration.
5. Acquire the ability to function effectively as part of a team to accomplish a set of common goals and objectives and efficiently communicate with project stakeholders.
6. Adopt a professional and ethical approach to decision making and related social responsibilities.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
CSIT111	Programming Fundamentals	6	
CSIT113	Problem Solving	6	
CSIT114	System Analysis	6	
CSIT115	Data Management & Security	6	CSIT111
CSIT121	Object Oriented Design & Programming	6	CSIT111
CSIT127	Networks and Communications	6	
CSIT128	Introduction to Web Technology	6	CSIT121
MATH221	Mathematics for Computer Science	6	
Year 2			
CSCI203	Algorithms & Data Structures	6	
CSIT214	IT Project Management	6	CSIT114
CSIT226	Human Computer Interaction	6	CSIT121
CSCI235	Database Systems	6	CSIT115 and CSIT121
CSCI251	Advanced Programming	6	CSIT121
Elective 1	200/300-LEVEL Comp. Science/IT	6	
Elective 2	200/300-LEVEL Comp. Science/IT	6	
Elective 3	Comp. Science/General Schedule	6	

Subject Code	Subject Name	CP	Pre-requisites
Year 3			
CSIT314	Software Development Methodologies	6	CSIT214 and 18cp at 200 level CSIT/ CSCI/ ISIT
Elective 4	300-LEVEL Comp. Science/IT	6	
Elective 5	300-LEVEL Comp. Science/IT	6	
Elective 6	Comp. Science/General Schedule	6	
Elective 7	Comp. Science/General Schedule	6	
Capstone			
CSIT321	Project	12	CSIT214 and 18cp, 200 level CSIT/CSCI/ISIT
General Education			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135 / GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101/ ARA 102	Arabic Language	6	

BACHELOR OF COMPUTER SCIENCE (CYBER SECURITY)

The Bachelor of Computer Science (Cyber Security) aims to prepare graduates to develop secure computers and computer networks, to ensure that the data stored and transmitted through them is protected from unauthorised access or to combat digital security threats and hazards. Moreover, with the advent of Internet of Things (IoT), both the public and private sectors are faced with new types of threats, new vulnerabilities, and hacking techniques and agents. IoT has hugely disrupted the traditional concept of digital security and introduced a more intricate concept: the concept of cybersecurity. There is a crucial need for talented cybersecurity professionals to address these issues. Skills derived from this major are in demand across all sectors: from banking to finance, insurance, retail, government and defence, all fields where there is sensitive data and information that is at risk of being hacked and breached via the internet.

Learning Outcomes

In addition to meeting the program learning outcomes of the Bachelor of Computer Science, students graduating from the Cyber Security major will further be able to:

- Apply the core fundamentals of cyber security to real world situations.
- Assess cyber security problems and vulnerabilities in software, computers, and networks and select appropriate security measures.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
CSIT111	Programming Fundamentals	6	
CSIT113	Problem Solving	6	
CSIT114	System Analysis	6	
CSIT115	Data Management & Security	6	CSIT111
CSIT121	Object Oriented Design & Programming	6	CSIT111
CSIT127	Networks and Communications	6	
CSIT128	Introduction to Web Technology	6	CSIT121
MATH221	Mathematics for Computer Science	6	
Year 2			
CSCI203	Algorithms & Data Structures	6	
CSIT214	IT Project Management	6	CSIT114
CSIT226	Human Computer Interaction	6	CSIT121
CSCI235	Database Systems	6	CSIT115 & CSIT121
CSCI251	Advanced Programming	6	CSIT121
CSCI262	System Security	6	CSIT111, CSIT115, CSIT127 & CSIT128
CSCI301	Contemporary Topics in Security	6	CSCI251
Elective 1	Comp. Science/General Schedule	6	
Year 3			
CSIT314	Software Development Methodologies	6	CSIT214 and 18cp @ 200 level CSIT/CSCI/ISIT
CSIT302	Cybersecurity	6	CSIT127 & 12cp at 100-level of CSIT
CSCI368	Network Security	6	CSCI251 & 12cp @ 200 level CSIT/CSCI
CSCI369	Ethical Hacking	6	CSCI251 and 12cp 200 level CSCI
Elective 2	Comp. Science/General Schedule	6	
Capstone			
CSIT321	Project	12	CSIT214 & 18cp, 200 level CSIT/CSCI/ISIT
General Education			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135/ GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101/ ARA 102	Arabic Language	6	

BACHELOR OF COMPUTER SCIENCE (GAME AND MOBILE DEVELOPMENT)

The Bachelor of Computer Science (Game and Mobile Development) aims to equip students with the necessary skills to apply techniques to the development of video games and related systems on computers and mobile platforms, including devices such as mobile phones, tablets and wearable devices. Students will be able to identify approaches to solving real-world problems in video games and multimedia systems. They will also develop both traditional computer science skills as well as creative skills including visualisation, interaction and communication skills. Experts in this field work in the entertainment industry, the game and mobile industry as mobile application developers, game programmers, animators, game engine developers, mobile entertainment and multimedia designers.

Learning Outcomes

In addition to meeting the program learning outcomes of the Bachelor of Computer Science, students graduating from the Multimedia and Game Development major will further be able to:

1. Apply the core fundamentals of the principles of design in animation, graphics, video, audio, web, and 2D/3D virtual worlds for designing games and mobile applications.
2. Develop, assess and integrate a range of software tools for creative games and mobile technology development.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
CSIT111	Programming Fundamentals	6	
CSIT113	Problem Solving	6	
CSIT114	System Analysis	6	
CSIT115	Data Management & Security	6	CSIT111
CSIT121	Object Oriented Design & Programming	6	CSIT111
CSIT127	Networks and Communications	6	
CSIT128	Introduction to Web Technology	6	CSIT121
MATH221	Mathematics for Computer Science	6	
Year 2			
CSCI203	Algorithms & Data Structures	6	
CSIT214	IT Project Management	6	CSIT114
CSIT226	Human Computer Interaction	6	CSIT121
CSCI235	Database Systems	6	CSIT115 and CSIT121
CSCI251	Advanced Programming	6	CSIT121
Major 1*	Select major subject from the list below	6	
Major 2*	Select major subject from the list below	6	
Elective 1	Comp. Science/General Schedule	6	
Year 3			
CSIT314	Software Development Methodologies	6	CSIT214 and 18cp @ 200 level CSIT/CSCI/ISIT
Major 3*	Select major subject from the list below	6	
Major 4*	Select major subject from the list below	6	
Major 5*	Select major subject from the list below	6	
Elective 2	Comp. Science/General Schedule	6	
Capstone			
CSIT321	Project	12	CSIT214 and 18cp, 200 level CSIT/CSCI/ISIT
*Plus 5 Major subjects (30cp) from the list below			
CSCI336	Interactive Computer Graphics	6	CSCI251 and 6cp of 200-level CSCI subjects
CSCI346	Game Development	6	CSCI236
CSCI356	Game Engine Essentials	6	CSIT121 and CSIT214
CSIT242	Mobile Applications Development	6	CSIT121
CSCI376	Multicore and GPU Programming	6	CSCI251 and 12cp @ 200 level CSIT/CSCI
General Education			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135 / GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101/ ARA 102	Arabic Language	6	

BACHELOR OF COMPUTER SCIENCE (BIG DATA)

The Bachelor of Computer Science (Big Data) aims to equip students with the necessary skills to apply techniques to the development of video games and related systems on computers and mobile platforms, including devices such as mobile phones, tablets and wearable devices. Students will be able to identify approaches to solving real-world problems in video games and multimedia systems. They will also develop both traditional computer science skills as well as creative skills including visualisation, interaction and communication skills. Experts in this field work in the entertainment industry, the game and mobile industry as mobile application developers, game programmers, animators, game engine developers, mobile entertainment and multimedia designers.

Learning Outcomes

In addition to meeting the program learning outcomes of the Bachelor of Computer Science, students graduating from the Multimedia and Game Development major will further be able to:

1. Apply the core fundamentals of the principles of big data, visualization and data analytics in real life applications.
2. Design and employ a range of algorithms, reports, and software tools for big data and data analytics.

Degree Structure

In addition to the Program Core and General Education subjects, students are required to complete the following 30 cp (5 subjects) major core subjects and 12cp (2 subjects) from the list of electives.

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
CSIT111	Programming Fundamentals	6	
CSIT113	Problem Solving	6	
CSIT114	System Analysis	6	
CSIT115	Data Management & Security	6	CSIT111
CSIT121	Object Oriented Design & Programming	6	CSIT111
CSIT127	Networks and Communications	6	
CSIT128	Introduction to Web Technology	6	CSIT121
MATH221	Mathematics for Computer Science	6	
Year 2			
CSCI203	Algorithms & Data Structures	6	
CSIT214	IT Project Management	6	CSIT114
CSIT226	Human Computer Interaction	6	CSIT121
CSCI235	Database Systems	6	CSIT115 and CSIT121
CSCI251	Advanced Programming	6	CSIT121
ISIT312	Big Data Management	6	CSIT115 and 24cp @ 100 level CSIT
CSCI316	Big Data Mining Techniques and Implementation	6	CSCI203 and MATH221
Elective 1	Comp. Science/General Schedule	6	
Year 3			
CSIT314	Software Development Methodologies	6	CSIT214 and 18 credit points at 200 level CSIT/CSCI/ISIT
CSCI317	Database Performance Tuning	6	CSCI235
CSCI323	Modern Artificial Intelligence	6	CSCI251 + 1 subject at 200 level - CSCI / CSIT
INFO411	Data Mining and Knowledge Discovery	6	36 cp
Elective 2	Comp. Science/General Schedule	6	
Capstone			
CSIT321	Project	12	CSIT214 and 18 credit points, 200 level CSIT/CSCI/ISIT
General Education			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135 / GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101/ ARA 102	Arabic Language	6	

List of Electives

Electives (18cp)			
Students need to choose 18cp (3 subjects) if they are not opting for any major of study			
CSCI236	3D Modelling and Animation	6	36 credit points at 100 level
CSCI262	System Security	6	CSIT111, CSIT115, CSIT127 & CSIT128
CSCI301	Contemporary Topics in Security	6	CSCI251
CSCI316	Big Data Mining Techniques and Implementation	6	CSCI203 and MATH221
CSCI317	Database Performance Tuning	6	CSCI235
CSCI323	Artificial Intelligence	6	CSCI251 + 1 subject at 200 level - CSCI / CSIT
CSCI336	Interactive Computer Graphics	6	CSCI251 and 6cp of 200-level CSCI subjects
CSCI346	Game Development	6	CSCI236
CSCI350	Internship	6	
CSCI356	Game Engine Fundamentals	6	CSIT121 and CSIT214
CSCI368	Network Security	6	CSCI251 & 12cp @ 200 level CSIT/CSCI
CSCI369	Ethical Hacking	6	CSCI251 and 12cp 200 level CSCI
CSCI370	Special Topics in Computer Science A	6	18 cp @ 200 level CSCI
CSCI376	Multicore and GPU Programming	6	CSCI251 and 12cp @ 200 level CSIT/CSCI
CSIT242	Mobile Applications Development	6	CSIT121
CSIT302	Cybersecurity	6	CSIT127 & 12cp at 100-level of CSIT
INE 301	Innovation & Entrepreneurship	6	
INFO411	Data Mining and Knowledge Discovery	6	36 cp
ISIT204	Principles of e-Business	6	24cp @ 100 level any subjects
ISIT224	Management Information Systems	6	24cp @ 100 level of any subject
ISIT301	Professional Practice & Ethics	6	18cp @ 200 level
ISIT312	Big Data Management	6	CSIT115 and 24cp @ 100 level CSIT
ACCY111	Accounting Fundamentals in Society	6	
ACCY112	Accounting in Organizations	6	ACCY111
ECON100	Economic Essentials for Business	6	
MARK101	Marketing Principles	6	
MARK301	Digital Marketing	6	MARK101
MARK343	International Marketing	6	MARK101
MGNT110	Introduction to Management	6	
MGNT201	Organizational Behaviour	6	MGNT110
MGNT206	Human Resource Management	6	MGNT110

BACHELOR OF BUSINESS INFORMATION SYSTEMS

Testamur Title of Degree:	Bachelor of Business Information Systems
Abbreviation:	BBIS
UOWD Course Code:	3164
Total Credit Points:	144 (plus additional GEC requirement subjects)
Duration:	3 Years (8 Sessions)
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

The Bachelor of Business Information Systems provides students with the knowledge and skills to apply appropriate technologies to achieve commercial advantage. The program teaches the methodologies and tools to harness information technology, from both strategic and operational perspectives, to create business efficiencies. It helps enhance analytical, problem solving, decision-making and critical thinking skills and helps to develop an understanding of key business concepts. The program offers graduates key skills required for any IT professional such as: organisational areas in IT, database management, programming, information systems analysis and design, communications and networks, security, eBusiness, project management, web-based technology, and professional practice in IT. Graduates of this program will be able to manage specifications, design and implement systems and select and apply appropriate technologies for the automation of business information.

Degree Requirements

To qualify for award of the degree of Bachelor of Business Information Systems, a candidate shall accrue an aggregate of at least 144 Credit Points (cp) which include 108cp of program core, 12cp of final year project and 24cp of General Electives. Students are also required to accrue an overall weighted average mark (WAM) of 50%.

Students need to complete 24cp (4 subjects) of electives from which 18cp must be CSCI-300 or ISIT-300 level subjects. Students can enrol in 6cp subject from the list of electives in computer science or business schedule as long as they meet the pre-requisite requirement. Student may not enrol in any 100-level subject as elective. Students are also advised to consult with their academic advisor about their elective enrolment.

Learning Outcomes

Upon successful completion of the Bachelor of Business Information Systems program, students will be able to:

1. Evaluate the strategic impact of information systems on organisational processes, profitability and decision-making effectiveness.
2. Develop specifications, design and implement information systems.
3. Manage the development and maintenance of business information systems.
4. Select and apply appropriate technologies for the automation of business information systems.
5. Acquire the ability to function effectively as part of a team to accomplish a set of common goals and objectives.
6. Adopt a professional and ethical approach to decision making and related social responsibilities.
7. Demonstrate an understanding of core knowledge of key business concepts.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
CSIT111	Programming Fundamentals	6	
CSIT113	Problem Solving	6	
CSIT114	Systems Analysis	6	
CSIT115	Data Management and Security	6	CSIT111
CSIT121	Object Oriented Design and Programming	6	CSIT111
CSIT127	Networks and Communications	6	
CSIT128	Introduction to Web Technology	6	CSIT121
MGNT110	Introduction to Management	6	
Year 2			
CSIT214	IT Project Management	6	CSIT114
ISIT224	Management Information System	6	24cp @ 100 level of any subject
ISIT204	Principles of eBusiness	6	24cp @100 level any subjects
CSIT226	Human Computer Interaction	6	CSIT111 & CSIT 113
ACCY111	Accounting Fundamentals in Society	6	
FIN 111	Introductory Business Finance	6	
MARK101	Marketing Principles	6	
MATH223	Mathematics for Information Technology	6	
Year 3			
CSIT314	Software Development Methodologies	6	CSIT214 and 18 credit points at 200 level CSIT/CSCI/ISIT
Elective 1	300-Level subject CSCI/CSIT	6	
Elective 2	300-Level subject CSCI/CSIT	6	
Elective 3	300-Level subject CSCI/CSIT	6	
Elective 4	Elective General	6	
Capstone			
CSIT321	Project	12	CSIT214 and 18 credit points, 200 level CSIT/CSCI/ISIT
General Education			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135 / GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101/ARA 102	Arabic Language	6	

BACHELOR OF ENGINEERING (HONOURS)

Testamur Title of Degree:	Bachelor of Engineering (Honours)
Abbreviation:	BE
Majors	Computer Engineering Electrical Engineering Telecommunications Engineering
UOWD Course Code:	3166
Total Credit Points:	192 (plus additional GEC requirement subjects)
Duration:	4 Years
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

The Bachelor of Engineering program with major in Computer Engineering, Electrical Engineering or Telecommunications Engineering trains engineers for exciting and successful careers as system designers and developers, engineering managers, analysts, educators and researchers. This program prepares graduates to be globally recognised professional engineers with the requisite knowledge, skills and attributes to further develop in their chosen careers. They will have the capacity to effectively lead and contribute to the advancement of engineering, technology and commerce. This program is designed to provide the basic theoretical background required by professional engineers as well as experience in working in teams and the ability to manage the process of designing and developing engineering solutions to address real world problems.

Degree Requirements

To qualify for award of the degree of Bachelor of Engineering in any of the specialisations, a candidate shall accrue an aggregate of at least 192 Credit Points (cp), which includes one general elective chosen from the general education subjects, in addition to two more general education subjects, for a total of 204 (cp). In addition, completes the professional experience subject ECTE399. Students are also required to accrue an overall weighted average mark (WAM) of 50%. This degree requirement is only applicable to students joining Autumn 2022 intake.

Learning Outcomes

Upon successful completion of the Bachelor of Engineering program, students will be able to:

- Demonstrate the ability to discern, interpret and evaluate information, and apply their learning in order to make decisions and articulate logical responses.
- Apply the acquired knowledge to practical problems and adopt a professional approach to decision making.
- Use all forms of expression to clearly and confidently communicate knowledge to others, appropriate to the context.
- Acquire the ability to function effectively as part of a team to accomplish a set of common goals and objectives.
- Initiate research and projects, and apply problem solving and critical thinking strategies to solve problems.
- Identify the strengths of individuals, societies and cultures that are both similar and dissimilar from one's own, with a professional view to improving existing relationships between community networks, workplaces and organisations.
- Demonstrate the ability to effectively gather, critically analyse, evaluate and interpret qualitative and quantitative information using computer technology.
- Initiate and apply problem-solving and critical thinking strategies to research, projects and organisational problems, including ethical dilemmas and accept responsibility for the solutions delivered.
- Demonstrate a sense of social commitment, responsibility and concern; feeling empowered to apply learning and show respect for citizens' rights to everyday situations.

ELECTRICAL ENGINEERING

Subject Code	Subject Name	CP	Pre-requisites
Flexible First Year			
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	MATH141
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
PHYS143	Physics for Engineers	6	
Year 2			
CSCI291	Programming for Engineers	6	ENGG100
ECTE202	Circuits and Systems	6	ENGG104 and MATH142
ECTE203	Digital Signal Processing	6	MATH142
ECTE212	Electronics	6	ECTE202
ECTE213	Engineering Electromagnetics	6	MATH142 and PHYS143
ECTE233	Digital Hardware	6	ENGG100 OR ENGG104
ECTE250	Engineering Design and Management 2	6	ECTE233 and ENGG100 and ENGG104
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
Year 3			
ECTE323	Power Engineering 2	6	ECTE324 and MATH283
ECTE324	Foundations in Electrical Energy Utilisation	6	ECTE202
ECTE333	Microcontroller Architecture and Applications	6	ECTE233 and CSCI291
ECTE344	Control Theory	6	ECTE 203
ECTE351	Engineering Design and Management 3	12	ENGG104 and ECTE250 and CSCI291 and ECTE233
ECTE363	Communication Systems	6	ECTE203 and MATH283
ECTE364	Data Communications	6	ENGG100 OR (MATH141 AND ECTE233)
ECTE399	Professional Experience	0	
Year 4			
ECTE498	Thesis A	12	Completion of 120cps
Select 18cp (3 subjects) from the following:			
ECTE423	Power System Analysis	6	ECTE323 and MATH283
ECTE433	Embedded Systems	6	ECTE333
ECTE465	Wireless Communication Systems	6	MATH283
ECTE412	Power Electronics and Drives	6	ECTE324 and MATH283
ECTE426	Power Distribution Systems	6	ECTE324 and MATH283
ECTE471	Robotics and Flexible Automation	6	MATH283
Electives			
Elective 1	From list of Technical Electives List	6	
Elective 2	From list of Technical Electives List	6	
Elective 3	General Elective from list of General Education Subjects	6	
General Education			
All student should have completed the following subjects where one is considered as a general elective			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135 / GEOG222 / PUBH115	UAE and International Relations / Society and Environment / Public Health	6	
ARA 101 / ARA 102	Arabic Language	6	

COMPUTER ENGINEERING

Subject Code	Subject Name	CP	Pre-requisites
Flexible First Year			
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	MATH141
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
PHYS143	Physics for Engineers	6	
Year 2			
CSCI291	Programming for Engineers	6	ENGG100
ECTE202	Circuits and Systems	6	ENGG104 and MATH142
ECTE203	Digital Signal Processing	6	MATH142
ECTE212	Electronics	6	ECTE202
ECTE213	Engineering Electromagnetics	6	MATH142 and PHYS143
ECTE233	Digital Hardware	6	ENGG100 OR ENGG104
ECTE250	Engineering Design and Management 2	6	ECTE233 and ENGG100 and ENGG104
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
Year 3			
ECTE324	Foundations in Electrical Energy Utilisation	6	ECTE202
ECTE331	Real-time Embedded Systems	6	ECTE233 and CSCI291
ECTE333	Microcontroller Architecture and Applications	6	ECTE233 and ENGG100
ECTE344	Control Theory	6	ECTE 203
ECTE351	Engineering Design and Management 3	12	ENGG104 and ECTE250 and CSCI291 and ECTE233
ECTE363	Communication Systems	6	ECTE203 and MATH283
ECTE364	Data Communications	6	ENGG100 OR (MATH141 AND ECTE233)
ECTE399	Professional Experience	0	
Year 4			
ECTE498	Thesis A	12	Completion of 120cps
ECTE477	Programming Autonomous Systems	6	CSCI291
Select 12cp (2 subjects) from the following:			
ECTE401	Multimedia Signal Processing	6	ECTE301
ECTE433	Embedded Systems	6	ECTE333
ECTE471	Robotics and Flexible Automation	6	MATH283
Electives			
Elective 1	From list of Technical electives	6	
Elective 2	From list of Technical electives	6	
Elective 3	General Elective from list of General Education subjects	6	

General Education (All student should have completed the following subjects where one is considered as a general elective)

ARTS317	Muslim Societies Across the Ages	6
IRAE135 / GEOG222 / PUBH115	UAE and International Relations / Society and Environment / Public Health	6
ARA 101 / ARA 102	Arabic Language	6

TELECOMMUNICATIONS ENGINEERING

Subject Code	Subject Name	CP	Pre-requisites
Year 1			
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	MATH141
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
PHYS143	Physics for Engineers	6	
Year 2			
CSCI291	Programming for Engineers	6	ENGG100
ECTE202	Circuits and Systems	6	ENGG104 and MATH142
ECTE203	Digital Signal Processing	6	MATH142
ECTE212	Electronics	6	ECTE202
ECTE213	Engineering Electromagnetics	6	MATH142 and PHYS143
ECTE233	Digital Hardware	6	ENGG100 OR ENGG104
ECTE250	Engineering Design and Management 2	6	ECTE233 and ENGG100 and ENGG104
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
Year 3			
ECTE324	Foundations in Electrical Energy Utilisation	6	ECTE202
ECTE331	Real-time Embedded Systems	6	ECTE233 and CSCI291
ECTE333	Microcontroller Architecture and Applications	6	ECTE233 and CSCI291
ECTE344	Control Theory	6	ECTE203
ECTE351	Engineering Design and Management 3	12	ENGG104 and ECTE250 and CSCI291 and ECTE233
ECTE363	Communication Systems	6	ECTE203 and MATH283
ECTE364	Data Communications	6	ENGG100 OR (MATH141 and ECTE233)
ECTE399	Professional Experience	0	
Year 4			
ECTE498	Thesis A	12	Completion of 120cps
ECTE469	Queuing Theory and Optimization	6	MATH283
Select 12cp (2 subjects) from the following:			
ECTE401	Multimedia Signal Processing	6	ECTE301
ECTE465	Wireless Communication Systems	6	MATH283
ECTE482	Network Engineering	6	ECTE364
Electives			
Elective 1	From list of Technical electives	6	
Elective 2	From list of Technical electives	6	
Elective 3	General Elective from list of General Education Subjects	6	

General Education (All student should have completed the following subjects which can be taken as general electives)

ARTS317	Muslim Societies Across the Ages	6
IRAE135 / GEOG222 / PUBH115	UAE and International Relations / Society and Environment / Public Health	6
ARA 101 / ARA 102	Arabic Language	6

List of Technical Subjects

Subject Code	Subject Title	Credit points	Pre – requisite
ECTE401	Multimedia Signal Processing	6	ECTE301
ECTE412	Power Electronics and Drives	6	ECTE324 and MATH283
ECTE423	Power System Analysis	6	ECTE323 and MATH283
ECTE426	Power Distribution Systems	6	ECTE324 and MATH283
ECTE433	Embedded Systems	6	ECTE333
ECTE465	Wireless Communication Systems	6	MATH283
ECTE469	Queuing Theory and Optimization	6	MATH283
ECTE471	Robotics and Flexible Automation	6	MATH283
ECTE474	Internet of Things	6	ECTE364 and CSCI291
ECTE477	Programming Autonomous Systems	6	CSCI291
INFO411	Data Mining and Knowledge Discovery	6	MATH283 and ECTE333

Testamur Title of Degree:	Bachelor of Engineering (Honours)
Abbreviation:	BE
Specialisations	Civil Engineering
UOWD Course Code:	3166
Total Credit Points:	192 (plus additional GEC requirement subjects)
Duration:	4 Years
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

The Bachelor of Engineering program with major in Civil Engineering trains engineers for exciting and successful careers as professional engineers with the requisite knowledge, skills and attitudes to further develop in their chosen careers. They have the capacity to effectively lead and contribute to the advancement of engineering, technology and commerce. The programs have been designed to provide the theoretical background required for professional engineers as well as experience in working in teams and the ability to manage the process of designing and developing engineering solutions to address real world problems.

Degree Requirements

To qualify for award of the degree of Bachelor of Engineering in any of the specializations, a candidate shall accrue an aggregate of at least 192 Credit Points (cp), which includes two general electives chosen from the general education subjects, in addition to one more general education subjects, for a total of 198 (cp). In addition, completes the professional experience subject ENGG454. Students are also required to accrue an overall weighted average mark (WAM) of 50%. This degree requirement is only applicable to students joining Autumn 2022 intake.

Learning Outcomes

Upon successful completion of the major in Civil Engineering, graduates of this major will be able to:

1. Demonstrate a comprehensive knowledge and understanding of the engineering principles underpinning structures, water engineering, construction materials, geotechnics, and surveying within the context of construction and detailed and conceptual multi-disciplinary design.
2. Identify, and predict the behaviour of building materials and utilise them appropriately and cost-effectively in construction.
3. Utilise sophisticated engineering analysis, software and design tools to simulate the real-world including computer aided design and modelling of engineering systems.
4. Apply fundamental concepts to estimate loadings, survey site conditions, and assess reliability in the design and performance of structures that comply with stipulated codes and standards.
5. Plan construction projects, taking into account environmental impact, and availability of building materials, machinery, and labour.
6. Evaluate the engineering properties of soils and rocks, and employ suitable ground management techniques to establish stable conditions for infrastructure and to mitigate natural hazards.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisites
Flexible First Year			
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	MATH141
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	
PHYS143	Physics for Engineers	6	
Year 2			
CHEM103	Introductory Chemistry for Engineers	6	
CIVL201	Computer Modelling for Civil Engineering	6	ENGG102
CIVL245	Construction Materials	6	
CIVL272	Surveying	6	
EESC252	Geology for Engineers 1	6	
ENGG251	Mechanics of Solids	6	ENGG102
ENGG252	Engineering Fluid Mechanics	6	MATH142
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
Year 3			
CIVL311	Structural Design 1	6	
CIVL314	Structural Design 2	6	CIVL311
CIVL322	Hydraulics and Hydrology	6	ENGG252
CIVL352	Structures 1	6	ENGG251
CIVL361	Geo-mechanics	6	
CIVL394	Construction	6	
Elective 1	Technical Elective 1	6	
Elective 2	General Elective 1	6	
Year 4			
CIVL444	Engineering Design	6	CIVL361, CIVL311 and CIVL322
CIVL458	Construction Management	6	
CIVL462	Principles of Foundation Engineering	6	CIVL361
ENGG452	Thesis A	12	Completion of 120cps, with a weighted average mark of 65 or above
ENGG454	Professional Experience	0	
ENGG461	Managing Engineering Projects	6	Completed at least 96cp
Elective 3	Technical Elective 2	6	
Elective 4	General Elective 2	6	
Technical Electives (12cp)			
CIVL463	Applied Geotechnical Engineering	6	
CIVL454	Structures 2	6	
CIVL489	Sustainable Road and Transport Engineering	6	
CIVL491	Applied Finite Element Analysis for Civil Engineers	6	

Civil Engineering Students are required to complete two technical electives (Technical Elective 1 and Technical Elective 2) and two general electives (General Elective 1 and General Elective 2) from faculty list of approved electives.

General Education (All student should have completed the following subjects which can be taken as general electives)

ARTS317	Muslim Societies Across the Ages	6
IRAE135/ GEOG222	UAE and International Relations / Society and Environment	6
ARA 101 / ARA 102	Arabic Language	6

BACHELOR OF ENGINEERING (HONOURS): MECHANICAL

Testamur Title of Degree:	Bachelor of Engineering (Honours)
Abbreviation:	BE
Specialisations	Mechanical Engineering
UOWD Course Code:	3166
Total Credit Points:	192 (plus additional GEC requirement subjects)
Duration:	4 Years
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

Mechanical Engineering has the broadest scope of all the branches of engineering, and graduates in this field have the core skills to adapt to other fields of engineering if they wish to do so. Our BEng (Hons) - Mechanical Engineering program is designed to provide the educational requirements which lead to professional mechanical engineers. The courses develop discipline-specific knowledge and understanding through the teaching of the core mechanical engineering subjects; engineering maths and computation, mechanics of materials, mechanics of machines, dynamics and vibrations, materials science and manufacturing, thermodynamics, fluid mechanics, CAD and engineering design

Degree Requirements

To qualify for award of the degree of Bachelor of Engineering in any of the specializations, a candidate shall accrue an aggregate of at least 192 Credit Points (cp), which includes two general electives chosen from the general education subjects, in addition to one more general education subjects, for a total of 198 (cp). In addition, completes the professional experience subject ENGG454 and the mechanical engineering workshop subject. Students are also required to accrue an overall weighted average mark (WAM) of 50%. This degree requirement is only applicable to students joining Autumn 2022 intake.

Learning Outcomes

Upon successful completion of the major in Mechanical Engineering, graduates will be able to:

- Demonstrate a comprehensive knowledge and understanding of the engineering principles underpinning machines, structures, process, systems, fluid dynamics, engineering materials, and detailed and conceptual multi-disciplinary design.
- Utilise sophisticated engineering analysis, software and design tools to simulate the real-world including computer aided design and modelling of engineering systems.
- Solve problems creatively by designing and managing the production of new and improved machines, systems and processes.
- Design, optimise and maintain machines, systems, and processes such as vehicles and engines; conventional and renewable energy production systems; manufacturing plant and machinery; bulk materials handling systems; building services, refrigeration and air conditioning systems; rail, road and aerospace systems.
- Measure and control the performance of machines and systems in the real-world using sensors and transducers, data acquisition systems, test facilities, and lab experimentation.
- Carry out innovative, conceptual and detailed design of systems and components by establishing key aspects of the problem, researching current knowledge, problem-solving, generating options and identifying feasible/optimal solutions.

Degree Structure

Subject Code	Subject Name	CP	Pre-requisite
Year 1			
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	MATH141
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	
PHYS143	Physics for Engineers	6	
Year 2			
CHEM103	Introductory Chemistry for Engineers	6	
MECH201	Engineering Analysis	6	MATH142
MECH203	Mechanical Workshop Practice	0	
MECH215	Mechanical Engineering Design 1	6	MECH203
MECH226	Machine Dynamics	6	ENGG102 and MATH142
MECH252	Thermodynamics, Experimental Methods and Analysis	6	
ENGG251	Mechanics of Solids	6	ENGG102
ENGG252	Engineering Fluid Mechanics	6	MATH142
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
Year 3			
ENGG378	Sustainable Energy Technologies	6	
MECH311	Mechanical Engineering Design 2	6	MECH215
MECH321	Dynamics of Engineering Systems	6	
MECH326	Dynamics of Mechanisms	6	MECH226
MECH341	Thermodynamics of Engineering Systems	6	MECH252
MECH343	Heat Transfer & Aerodynamics	6	ENGG252
MECH365	Control of Machines and Processes	6	MECH321
MECH382	Manufacturing Engineering Principles	6	ENGG103
Year 4			
ENGG452	Thesis A	12	Completion of 120cps, with a weighted average mark of 65 or above
ENGG454	Professional Experience	0	
ENGG461	Managing Engineering Projects	6	Completed at least 96 credit points
MECH419	Finite Element Methods in Engineering	6	ENGG251 and MECH201
Electives (24cp)			
<i>Mechanical Engineering Students are required to complete two technical electives (Technical Elective 1 and Technical Elective 2) and two general electives (General Elective 1 and General Elective 2) from faculty list of approved electives.</i>			
Elective 1	Technical Elective 1	6	
Elective 2	Technical Elective 2	6	
Elective 3	Selected General Education Elective 1	6	
Elective 4	Selected General Education Elective 2	6	
Technical Electives (12cp)			
CIVL311	Structural Design 1	6	
ECTE471	Robotics and Flexible Automation	6	MATH283
MECH431	Computational Fluid Dynamics	6	
MECH474	Reliability Engineering	6	
MECH482	Special Topics in Mechanical Engineering 2	6	
MECH489	Engineering Asset Management	6	
General Education (All student should have completed the following subjects which can be taken as general electives)			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135/ GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101 / ARA 102	Arabic Language	6	

BACHELOR OF ENGINEERING (HONOURS): MECHATRONIC

Testamur Title of Degree:	Bachelor of Engineering (Honours)
Abbreviation:	BE
Specialisations	Mechatronic Engineering
UOWD Course Code:	3166
Total Credit Points:	192 (plus additional GEC requirement subjects)
Duration:	4 Years
QF Emirates Level	7
Home Faculty:	Faculty of Engineering and Information Sciences
Intake Session(s):	Autumn/Spring/Winter
Delivery Mode:	On Campus (Face-to-face)

Mechatronics, as a multidisciplinary engineering discipline, develops the skill sets demanded for in areas such as Industry 4.0 systems, Internet of Things, Robotic Systems, drones, Autonomous Vehicles, Assistive technologies, Medical Appliances and Robots. It focuses on mechanical, electrical and electronic, computer engineering systems, and their interdisciplinary combination in order to create smarter systems of great use to us. Industrial robots are a famous example of mechatronic engineering: computer-controlled mechanical arms capable of performing manual labour with greater strength, speed and efficiency than a human. The reality is computer control of systems is so commonplace in homes, business and industry that nearly everyone uses a mechatronic system every day, from washing machines to central air-conditioning, cameras and computer printers.

In the Bachelor of Engineering (Honours) – Mechatronic Engineering Degree you will study key areas and develop skills in Mechanical Engineering, Electrical Engineering, Computer Engineering, software and hardware design in order to be able to design, develop and manage complex mechatronic systems. The course provides essential foundations for dealing with devices such as Sensors, Actuators, Control Systems, Embedded Systems, Robotics, Industrial hydraulics and Pneumatics, Industrial Automation, Programmable Logic Controllers (PLC), Microcontrollers, Cyber-Physical Systems, 3D Printing, and Integrated Technologies such as AI and IoT.

Degree Requirements

To qualify for award of the degree of Bachelor of Engineering in any of the specializations, a candidate shall accrue an aggregate of at least 192 Credit Points (cp), which includes two general electives chosen from the general education subjects, in addition to one more general education subjects, for a total of 198 (cp). In addition, completes the professional experience subject ENGG454 and the mechanical engineering workshop subject. Students are also required to accrue an overall weighted average mark (WAM) of 50%. This degree requirement is only applicable to students joining Autumn 2022 intake.

Learning Outcomes

Upon successful completion of the major in Mechatronic Engineering, graduates will be able to:

- Demonstrates the ability to discern, interpret and evaluate disciplinary literature independently, to support an argument or strategy.
- Apply Engineering knowledge & Skills to practical problems and adopt a professional approach to decision making.
- Communicate clearly and coherently in writing to a range of audiences, with an ability to integrate knowledge, research, data, analysis and critical evaluation.
- Communicate verbally to a range of audiences using appropriate language in presentations, consultation and negotiation
- Work productively as part of a team with the capacity for leadership, recognizing the roles, responsibilities and accountabilities of team members.
- Initiate research and projects, and apply problem solving and critical thinking strategies to develop innovative solutions to complex engineering problems.
- Show respect for the views, values and culture of others in settings involving a range of stakeholders and consider alternate perspectives in design and project management
- Appreciate the importance of sustainable engineering design, and seek to maximize positive social and environmental outcomes in engineering design, practice and development

Degree Structure

Subject Code	Subject Name	CP	Pre-requisite
Year 1			
ENGG100	Engineering Computing and Analysis	6	
ENGG102	Fundamentals of Engineering Mechanics	6	
ENGG103	Materials in Design	6	
ENGG104	Electrical Systems	6	
ENGG105	Engineering Design for Sustainability	6	
MATH141	Foundations of Engineering Mathematics	6	
MATH142	Essentials of Engineering Mathematics	6	MATH141
PHYS143	Physics for Engineers	6	
Year 2			
CSCI291	Programming for Engineers	6	ENGG100
ECTE203	Signals and Systems	6	MATH142
ECTE233	Digital Hardware	6	ENGG100 OR ENGG104
ENGG251	Mechanics of Solids	6	ENGG102
MATH283	Advanced Engineering Mathematics and Statistics	6	MATH142
MECH203	Mechanical Workshop Practice	0	
MECH215	Mechanical Engineering Design 1	6	MECH203
MECH226	Machine Dynamics	6	ENGG102 and MATH142
MECH252	Thermodynamics, Experimental Methods and Analysis	6	
Year 3			
ECTE202	Circuits and Systems	6	ENGG104 and MATH142
ECTE212	Electronics	6	ECTE202
ECTE333	Microcontroller Architecture and Applications	6	ECTE233 and CSCI291
ECTE344	Control Theory	6	ECTE203
ECTE351	Engineering Design and Management 3	12	ENGG104 and ECTE250 and CSCI291 and ECTE233
ENGG252	Engineering Fluid Mechanics	6	MATH142
MECH311	Mechanical Engineering Design 2	6	MECH215
MECH382	Manufacturing Engineering Principles	6	ENGG103
Year 4			
ECTE333	Microcontroller Architecture and Applications	6	ECTE233 and CSCI291
ECTE471	Robotics and Flexible Automation	6	MATH283
ECTE498	Thesis A	12	Completion of 120 cps
ENGG454	Professional Experience	0	
ENGG461	Managing Engineering Projects	6	Completed at least 96 credit points
MECH470	Applied topics in Mechatronics	6	MECH203
	Technical Elective	6	
	General Elective	6	
Technical Electives (6 cp)			
ECTE363	Communication Systems	6	ECTE203 and MATH283
ECTE401	Multimedia Signal Processing	6	ECTE203
ECTE433	Embedded Systems	6	ECTE333
ECTE465	Wireless Communication Systems	6	MATH283
ECTE474	Internet of Things	6	ECTE364 and CSCI291
ECTE477	Programming Autonomous Systems	6	CSCI291
General Education (All student should have completed the following subjects which can be taken as general electives)			
ARTS317	Muslim Societies Across the Ages	6	
IRAE135/ GEOG222	UAE and International Relations / Society and Environment	6	
ARA 101 / ARA 102	Arabic Language	6	

CLASSIFICATION OF HONOURS DEGREE FOR ENGINEERING PROGRAMS

WAM Range	Classification Awarded
77.5%-100%	Class I
72.5%-77.5%	Class II, Division 1
67.5%-72.5%	Class II, Division 2
62.5%-67.5%	Class III
50% - 62.5%	Honours with no classification